

Discrete Time Model (EC331 Draft Model - Last Update 1 December 2025)

A Two-Capital Growth Model with Housing Frictions and Spatial Productivity

1 ECONOMIC ENVIRONMENT

1.1 Time, Population, and Technology

Time is discrete, $t = 0, 1, 2, \dots$ (periods measured in years).

Definition 1.1 (Population). Total population evolves according to:

$$L_{t+1} = (1 + n)L_t, \quad L_0 > 0 \text{ given}, \quad n \geq 0 \quad (1)$$

where n is the exogenous population growth rate.

Definition 1.2 (Frontier Technology). The frontier productivity level evolves endogenously according to:

$$A_{0,t+1} = (1 + g(s_t)) A_{0,t}, \quad A_{0,0} > 0 \text{ given} \quad (2)$$

where $g(s_t)$ is the endogenous growth rate of frontier technology.

Definition 1.3 (Effective Labor). Effective labor units in period t are:

$$E_t \equiv A_{0,t} L_t \quad (3)$$

Remark 1.1. Effective labor grows at gross rate $(1 + g(s_t))(1 + n)$, consistent with labor-augmenting (Harrod-neutral) technological change with endogenous productivity growth.

1.2 Spatial Structure

The economy consists of a continuum of locations indexed by $\ell \in [0, 1]$.

Definition 1.4 (Location Types). Each location ℓ is characterized by a **base productivity** $\omega(\ell) \in \{\omega_R, \omega_U\}$ where:

- $\omega_U > 1$: High-productivity (urban) locations
- $\omega_R = 1$: Low-productivity (rural) locations (normalization)

Assumption 1.1 (Spatial Distribution). The measure of high-productivity locations is $\mu_U \in (0, 1)$. We assume $\mu_U > \pi$ (abundant urban land relative to households who can access it).

1.3 Household Heterogeneity in Housing Access

The population is partitioned into two types based on **ability to access urban housing markets**.

Definition 1.5 (Household Types). • **Type π ("Unconstrained")**: Share $\pi \in (0, 1)$ of population; possess sufficient wealth, credit access, or legal status to participate in urban housing markets.

- **Type $(1 - \pi)$ ("Constrained")**: Share $1 - \pi$ of population; face binding constraints preventing urban housing access.

Assumption 1.2 (Exogenous Type Distribution). The share π is a structural parameter reflecting credit market institutions, wealth distribution, and legal/regulatory barriers. It is exogenous to the growth process.

Calibration note: Empirical estimates of π will be derived from homeownership rates and rental market participation data in future empirical work.

2 PRODUCTION TECHNOLOGY

2.1 Aggregate Production Function

Definition 2.1 (Technology). Output in period t is produced via:

$$Y_t = K_t^\alpha H_t^\beta (A_{\text{eff},t} L_t)^{1-\alpha-\beta} \quad (4)$$

where:

- $K_t \geq 0$: **Business capital** stock
- $H_t \geq 0$: **Quality-adjusted residential capital** stock (defined below)
- $A_{\text{eff},t} > 0$: **Effective total factor productivity** (endogenous)
- $L_t > 0$: Labor force
- $\alpha, \beta \in (0, 1)$: Output elasticities
- $\alpha + \beta < 1$: Ensures decreasing returns to reproducible capital

Assumption 2.1 (Production Function Properties). 1. Constant returns to scale in (K, H, AL)

2. Cobb-Douglas form (tractability; consistent with balanced growth)

3. Factor complementarity: All inputs essential

Calibration note: Parameters α and β will be calibrated to match national accounts data on capital income shares.

2.2 Quality-Adjusted Housing Capital

Definition 2.2 (Quality-Adjusted Residential Capital). Let $\tilde{H}_t$ denote the **physical stock** of residential structures (measured in square meters or housing units). Let q_t denote the **average quality** of the housing stock (embodying appliances, insulation, connectivity, safety features).

We define **quality-adjusted residential capital** as:

$$H_t \equiv \frac{\tilde{H}_t \cdot q_t}{A_{0,t}} \quad (5)$$

This normalization ensures that H_t is measured in **efficiency units**: housing of constant quality relative to the technological frontier.

Assumption 2.2 (Quality Standards Track Technology). Along a balanced growth path, the average quality of newly constructed housing rises proportionally with aggregate productivity:

$$q_t = \kappa \cdot A_{0,t} \quad (6)$$

where $\kappa > 0$ is a constant reflecting construction standards.

Implication: The definition $H_t = \tilde{H}_t q_t / A_{0,t} = \kappa \tilde{H}_t$ implies that our effective housing stock is a *quality-adjusted* measure of physical structures. Along a balanced growth path (BGP), the intensive housing variable

$$h_t \equiv \frac{H_t}{A_{0,t} L_t}$$

is stationary, while the level of housing H_t grows at the same gross rate as output and other reproducible aggregates.

Remark 2.1. From this point forward, all equilibrium conditions are written in terms of H_t and h_t . Whenever we speak of “stationarity” of housing on a BGP, we refer to the intensive quantity h_t , not the level H_t . The hedonic equilibrium derivation of the quality scaling law is provided in **Appendix A**.

Interpretation: H_t is residential capital measured in constant-quality units, while h_t is residential capital per effective worker. Just as GDP is deflated by a price index, housing capital is deflated by a quality index $A_{0,t}$ and scaled by population.

2.3 Effective TFP and Spatial Allocation

Effective TFP depends on the **spatial distribution of labor** across location types.

Definition 2.3 (Spatial Allocation). Let $s_t \in [0, \pi]$ denote the **share of total population** residing in high-productivity (urban) locations at time t .

Definition 2.4 (Effective TFP).

$$A_{\text{eff},t} = A_{0,t} \cdot \Omega(s_t) \quad (7)$$

where:

$$\Omega(s) = [s \cdot \omega_U + (1 - s) \cdot \omega_R]^\psi \quad (8)$$

with:

- $\omega_U > \omega_R = 1$: Productivity differential
- $\psi > 0$: **Agglomeration elasticity**

Assumption 2.3 (Functional Form Justification). This reduced-form agglomeration function captures three key mechanisms documented in urban economics:

1. **Knowledge spillovers:** Dense urban environments facilitate face-to-face interaction, idea exchange, and learning-by-doing (Glaeser 1999; Moretti 2004).
2. **Thick labor markets:** Larger cities enable better worker-firm matching and specialization (Duranton & Puga 2004).
3. **Input sharing:** Urban agglomeration reduces transaction costs and enables specialized intermediate inputs (Marshall 1890; Combes & Gobillon 2015).

The power function form is consistent with spatial equilibrium models (Rossi-Hansberg 2005) and has strong empirical support.

Properties:

- $\Omega(0) = \omega_R^\psi = 1$ (rural productivity is normalized)
- $\Omega(\pi) = [\pi\omega_U + (1 - \pi)]^\psi$ (maximum urbanization)
- $\Omega'(s) = \psi[s\omega_U + (1 - s)]^{\psi-1}(\omega_U - 1) > 0$ (monotonically increasing)
- Ω is continuous and bounded on $[0, \pi]$

Calibration note: Parameters ω_U and ψ will be estimated from cross-city wage premia and agglomeration elasticity studies in the empirical section.

3 HOUSING MARKET FRICTIONS

Urban housing is rationed due to search-and-matching frictions.

3.1 Housing Market States and Matching

Definition 3.1 (Housing Market States). At the beginning of period t :

- H_t^{occ} : Measure of occupied housing units (households currently residing)
- H_t^{vac} : Measure of vacant units available for matching
- B_t : Measure of households searching for urban housing

Definition 3.2 (Matching Function). The measure of successful matches in period t is:

$$M_t = \chi (H_t^{\text{vac}})^\mu B_t^{1-\mu} \quad (9)$$

where:

- $\chi > 0$: **Matching efficiency**
- $\mu \in (0, 1)$: **Matching elasticity** w.r.t. vacancies

Assumption 3.1 (Cobb-Douglas Matching). Standard in search literature (Petrongolo & Pissarides 2001).

3.2 Vacancy Dynamics (Endogenous Steady-State Vacancy Rate)

Definition 3.3 (Separation and Depreciation). • $\sigma \in (0, 1)$: **Separation rate** (fraction of occupied units vacated per period due to household relocation)

- $\delta_v \in (0, 1)$: **Vacancy depreciation** (fraction of vacant units that deteriorate and exit the stock)

Laws of Motion (Vacancy Block):

$$\boxed{H_{t+1}^{\text{occ}} = (1 - \sigma)H_t^{\text{occ}} + M_t} \quad (10)$$

$$\boxed{H_{t+1}^{\text{vac}} = (1 - \delta_v)H_t^{\text{vac}} + \sigma H_t^{\text{occ}} - M_t} \quad (11)$$

$$\boxed{B_{t+1} = \pi L_{t+1} - H_{t+1}^{\text{occ}}} \quad (12)$$

Remark 3.1. These three equations form a **fast subsystem** (vacancy dynamics adjust quickly relative to capital accumulation). The derivation of the stationary vacancy rate and stability analysis are provided in **Appendix B**.

Proposition 3.1 (Stationary Vacancy Rate). *Given parameters $(\chi, \mu, \sigma, \delta_v)$ and a slow-moving total stock H_t , the vacancy block converges quickly to a unique stationary ratio:*

$$v^* \equiv \frac{H^{\text{vac}}}{H} = \frac{\sigma}{1 + \sigma - \delta_v} \quad (13)$$

Proof: See **Appendix B**.

Calibration note: Parameters σ and δ_v will be estimated from housing turnover data and vacancy rate statistics in future empirical work.

3.3 Reduced-Form Urban Occupancy

Definition 3.4 (Urban Occupancy). The share of total population residing in urban areas is:

$$s_t \equiv \frac{H_t^{\text{occ}}}{L_t} \quad (14)$$

Proposition 3.2 (Implicit Urban Occupancy Function). *In quasi-steady-state with $H_t^{\text{vac}} = v^* H_t$ and $B_t = \pi L_t - s_t L_t$, the urban occupancy satisfies the implicit equation:*

$$\boxed{s_t = \Gamma(h_t)} \quad (15)$$

defined by:

$$\boxed{s_t^{1/(1-\mu)}(\pi - s_t) = \tilde{\chi} h_t^{\mu/(1-\mu)}} \quad (16)$$

where $h_t \equiv H_t/(A_{0,t}L_t)$ is housing capital per effective worker, and:

$$\tilde{\chi} \equiv \left(\frac{\chi(v^*)^\mu}{\sigma} \right)^{1/(1-\mu)} \quad (17)$$

Proposition 3.3 (Properties of Γ). *The function $\Gamma : \mathbb{R}_+ \rightarrow [0, \pi]$ satisfies:*

1. **Domain and range:** $\Gamma(h) \in [0, \pi]$ for all $h \geq 0$
2. **Boundary conditions:** $\Gamma(0) = 0$ and $\lim_{h \rightarrow \infty} \Gamma(h) = \pi$
3. **Monotonicity:** $\Gamma'(h) > 0$ for all $h > 0$

4. **Continuity:** Γ is continuous on $\mathbb{R}_+$

Proof: See **Appendix C** for the detailed derivation and monotonicity verification.

Numerical Solution: For given h_t , solve the implicit equation for $s_t \in [0, \pi]$ using Newton-Raphson iteration or bisection.

Remark 3.2. The main text uses **only** the implicit equation. A low-congestion approximation ($s_t \approx \chi_{\text{eff}} h_t^\mu$ when $s_t \ll \pi$) is discussed in **Appendix C** for intuition but is not used in the model solution.

4 ENDOGENOUS TECHNOLOGICAL PROGRESS

4.1 Innovation and Urban Agglomeration

Definition 4.1 (Endogenous Growth Function). The growth rate of frontier technology depends on urban concentration:

$$g(s_t) = g_0 + \phi \cdot s_t^\zeta \quad (18)$$

where:

- $g_0 \geq 0$: **Baseline growth rate** (exogenous component from global knowledge diffusion)
- $\phi > 0$: **Innovation elasticity** (sensitivity of frontier growth to urban concentration)
- $\zeta > 0$: **Curvature parameter** (returns to scale in idea production)

Assumption 4.1 (Microfoundation). This reduced form captures three mechanisms linking urbanization to innovation:

1. **Direct innovation:** Cities host universities, R&D labs, and patent-intensive firms (Bloom, Jones, Van Reenen & Webb 2020).
2. **Knowledge spillovers:** Face-to-face contact accelerates idea recombination and technology adoption (Jones 1995; Rossi-Hansberg, Sarte & Schwartzman 2019).
3. **Human capital externalities:** Urban density raises skill acquisition through learning-by-doing and mentorship (Lucas 1988; Moretti 2004).

Assumption 4.2 (Boundedness). To ensure the intensive-form system admits a stationary BGP, we require:

$$g(s) \in [g_0, g_{\max}] \quad (19)$$

for some $g_{\max} < \infty$. This is satisfied if:

- $\zeta \leq 1$ (diminishing returns to urban concentration), **OR**
- ϕ is sufficiently small such that $g_0 + \phi\pi^\zeta < g_{\max}$

Calibration note: Parameters g_0 , ϕ , and ζ will be estimated from cross-city patent data and innovation elasticity studies in future empirical work.

5 INVESTMENT ALLOCATION: PORTFOLIO CHOICE WITH ADJUSTMENT COSTS

5.1 Rental Rates and Returns

Definition 5.1 (Rental Rates). Competitive firms rent capital at rates equal to marginal products:

$$r_{K,t} = \alpha \frac{Y_t}{K_t} \quad (20)$$

$$r_{H,t} = \beta \frac{Y_t}{H_t} + \rho_t \quad (21)$$

where ρ_t is the **congestion rent** capturing the marginal value of alleviating housing scarcity.

Definition 5.2 (Congestion Rent). Housing in urban areas generates a positive externality by enabling more workers to access high-productivity jobs. The shadow value is derived from the impact of H_t on effective TFP:

$$\rho_t = (1 - \alpha - \beta) \frac{Y_t}{A_{\text{eff},t} L_t} \frac{\partial \Omega(s_t)}{\partial s_t} \frac{\partial s_t}{\partial H_t} \quad (22)$$

Derivation: See **Appendix D** for the complete implicit-function derivative calculation.

Linear Approximation: Near a steady state, linearizing around s^* :

$$\rho_t \approx \rho_0 + \rho_1(\pi - s_t) \quad (23)$$

where $\rho_1 > 0$ depends on the structural parameters $(\psi, \omega_U, \mu, \pi)$.

5.2 Investor Problem with Adjustment Costs

Assumption 5.1 (Representative Investor). A representative investor allocates total saving $I_t = \varsigma Y_t$ between business capital and housing to maximize expected returns subject to quadratic adjustment costs.

Notation change: To avoid confusion with occupancy s_t , we denote the saving rate by $\varsigma \in (0, 1)$.

Problem:

$$\max_{\theta_{H,t}} \mathbb{E}_t [(1 - \theta_{H,t}) R_{K,t+1} + \theta_{H,t} R_{H,t+1}] - \frac{\varphi}{2} (\theta_{H,t} - \bar{\theta})^2 \quad (24)$$

where:

- $R_{K,t+1} = r_{K,t} - \delta_K$: Net return to business capital
- $R_{H,t+1} = r_{H,t} - \delta_H$: Net return to housing capital
- $\theta_{H,t} \in [0, 1]$: Share of total investment allocated to housing
- $\varphi > 0$: **Adjustment cost parameter**
- $\bar{\theta} \in (0, 1)$: **Long-run housing investment share** (when $s_t = \pi$, no congestion)

Assumption 5.2 (Adjustment Cost Justification). The quadratic cost function $\frac{\varphi}{2} (\theta_{H,t} - \bar{\theta})^2$ captures:

1. **Regulatory frictions:** Zoning laws, building permits, environmental reviews slow reallocation.

2. **Sector-specific capital:** Construction equipment, skilled trades are costly to reallocate between sectors.
3. **Coordination costs:** Large-scale shifts between residential and commercial construction require time-intensive coordination.

The quadratic form ensures interior solutions and smooth dynamics. The parameter φ governs the speed of portfolio adjustment.

First-Order Condition:

$$\mathbb{E}_t[R_{H,t+1} - R_{K,t+1}] = \varphi(\theta_{H,t} - \bar{\theta}) \quad (25)$$

$$\boxed{\theta_{H,t} = \bar{\theta} + \frac{1}{\varphi} \mathbb{E}_t[R_{H,t+1} - R_{K,t+1}]} \quad (26)$$

Proposition 5.1 (Investment Allocation Rule). *Under the linear approximation for congestion rent and assuming steady-state factor shares:*

$$\boxed{\theta_{H,t} = \bar{\theta} + \eta(\pi - s_t)} \quad (27)$$

where:

$$\boxed{\eta \equiv \frac{\rho_1}{\varphi}} \quad (28)$$

is the *policy responsiveness parameter*.

Interpretation:

- **Numerator ρ_1 :** Marginal value of housing in reducing congestion
- **Denominator φ :** Cost of adjusting portfolio
- Higher congestion ($\pi - s_t$ large) $\rightarrow$ tilt toward housing investment
- Higher adjustment costs $\rightarrow$ slower response

Boundary conditions: We impose:

$$\theta_{H,t} \in [\theta_{\min}, \theta_{\max}] \subset (0, 1) \quad (29)$$

to ensure feasible allocations.

5.3 Empirical Identification of η

Identification challenges: Separately identifying η , $\Omega(s)$, and $g(s)$ from aggregate or city-level data requires:

1. **Independent variation:** Need exogenous shocks to housing supply (e.g., land-use regulatory changes, natural experiments) that affect s_t but not directly g_t .
2. **Multiple data sources:**
 - Occupancy rates s_t from census/administrative data
 - Agglomeration effects $\Omega(s)$ from cross-city wage regressions
 - Innovation rates $g(s)$ from patent data or TFP growth
3. **Institutional variation:** Cross-country or cross-region differences in adjustment costs φ (zoning stringency, construction regulations).

Proposed empirical strategy: Panel regression of residential investment shares on housing market tightness indicators:

$$\text{ResInvShare}_{i,t} = \alpha_i + \beta \cdot \text{TightnessProxy}_{i,t} + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t} \quad (30)$$

where $\text{TightnessProxy}_{i,t}$ captures $(\pi - s_t)$ through:

- House price index deviations from trend
- Price-to-rent ratios
- Vacancy rate deviations

Controls $\mathbf{X}_{i,t}$ include GDP growth, interest rates, demographic changes, and regulatory stringency indices.

Calibration note: This regression will be estimated in future empirical work using OECD panel data. The coefficient $\hat{\beta}$ provides an estimate of η .

6 CAPITAL ACCUMULATION

6.1 Resource Constraint

$$Y_t = C_t + I_{K,t} + I_{H,t} \quad (31)$$

Assumption 6.1 (Solow Saving Rule).

$$I_t = \varsigma Y_t, \quad C_t = (1 - \varsigma)Y_t \quad (32)$$

where $\varsigma \in (0, 1)$ is the aggregate saving rate (reduced-form parameter).

Calibration note: The saving rate ς will be set to match aggregate investment-to-GDP ratios in the data.

6.2 Investment Allocation

$$I_{K,t} = [1 - \theta_{H,t}] \varsigma Y_t, \quad I_{H,t} = \theta_{H,t} \varsigma Y_t \quad (33)$$

where:

$$\theta_{H,t} = \bar{\theta} + \eta(\pi - s_t) \quad (34)$$

6.3 Laws of Motion

Definition 6.1 (Capital Accumulation).

$$\boxed{K_{t+1} = (1 - \delta_K)K_t + I_{K,t}} \quad (35)$$

$$\boxed{H_{t+1} = (1 - \delta_H)H_t + I_{H,t}} \quad (36)$$

where:

- δ_K : Business capital depreciation
- δ_H : **Effective depreciation** of quality-adjusted housing (includes physical depreciation and quality obsolescence)

Remark 6.1. Since H_t is measured in efficiency units, δ_H captures both physical wear and the rate at which existing structures fall below acceptable quality standards.

Calibration note: Depreciation rates δ_K and δ_H will be calibrated to match observed capital stock dynamics in national accounts data.

7 THE COMPLETE DYNAMICAL SYSTEM

7.1 Intensive Form Variables

$$k_t \equiv \frac{K_t}{A_{0,t}L_t}, \quad h_t \equiv \frac{H_t}{A_{0,t}L_t}, \quad y_t \equiv \frac{Y_t}{A_{0,t}L_t} \quad (37)$$

7.2 Production (Intensive Form)

$$y_t = k_t^\alpha h_t^\beta \Omega(s_t)^{1-\alpha-\beta} \quad (38)$$

where:

$$\Omega(s_t) = [s_t \omega_U + (1 - s_t)]^\psi \quad (39)$$

7.3 Urban Occupancy (Implicit Equation)

$$s_t = \Gamma(h_t) \quad (40)$$

defined implicitly by:

$$s_t^{1/(1-\mu)}(\pi - s_t) = \tilde{\chi} h_t^{\mu/(1-\mu)} \quad (41)$$

Numerical Solution: For given h_t , solve this equation for $s_t \in [0, \pi]$ using Newton's method or bisection.

7.4 Investment Allocation

$$\theta_{H,t} = \min\{\max\{\bar{\theta} + \eta(\pi - s_t), \theta_{\min}\}, \theta_{\max}\} \quad (42)$$

$$\theta_{K,t} = 1 - \theta_{H,t} \quad (43)$$

7.5 Laws of Motion (Intensive Form)

$$k_{t+1} = \frac{(1 - \delta_K)k_t + [1 - \theta_{H,t}]\varsigma y_t}{(1 + g(s_t))(1 + n)} \quad (44)$$

$$h_{t+1} = \frac{(1 - \delta_H)h_t + \theta_{H,t}\varsigma y_t}{(1 + g(s_t))(1 + n)} \quad (45)$$

where:

$$g(s_t) = g_0 + \phi s_t^\zeta \quad (46)$$

This is a two-dimensional autonomous difference equation system in (k_t, h_t) with endogenous growth.

8 BALANCED GROWTH PATH

8.1 Definition

Definition 8.1 (Balanced Growth Path). A BGP is a state (k^*, h^*) such that:

$$k_{t+1} = k_t = k^*, \quad h_{t+1} = h_t = h^* \quad \forall t \quad (47)$$

with $s^* = \Gamma(h^*)$ and $g^* = g(s^*)$ constant.

Along a BGP:

- Intensive variables (k^*, h^*, y^*, s^*) are constant
- Level variables (K_t, H_t, Y_t) grow at rate $g^* + n + g^*n$
- Frontier technology $A_{0,t}$ grows at rate g^*

Lemma 8.1 (Balanced-Growth Scaling). *Consider any path $(K_t, H_t, Y_t, A_{0,t}, L_t)$ such that:*

1. *Population and frontier technology evolve as*

$$L_{t+1} = (1+n)L_t, \quad A_{0,t+1} = (1+g^*)A_{0,t},$$

with constant $n \geq 0$ and $g^ \geq 0$.*

2. *Intensive capital stocks are constant:*

$$k_t \equiv \frac{K_t}{A_{0,t}L_t} = k^*, \quad h_t \equiv \frac{H_t}{A_{0,t}L_t} = h^*.$$

3. *Output satisfies*

$$Y_t = K_t^\alpha H_t^\beta (A_{0,t}\Omega(s^*)L_t)^{1-\alpha-\beta},$$

with $s_t = s^$ and $\Omega(s^*)$ constant.*

Then:

1. *The level variables K_t, H_t, Y_t all grow at the common gross rate*

$$(1+g^*)(1+n),$$

i.e.

$$\frac{K_{t+1}}{K_t} = \frac{H_{t+1}}{H_t} = \frac{Y_{t+1}}{Y_t} = (1+g^*)(1+n).$$

2. *The intensive output $y_t \equiv Y_t/(A_{0,t}L_t)$ is constant at*

$$y^* = (k^*)^\alpha (h^*)^\beta \Omega(s^*)^{1-\alpha-\beta}.$$

**Proof:* Under the stated conditions,*

$$K_t = k^* A_{0,t} L_t, \quad H_t = h^* A_{0,t} L_t.$$

Using $A_{0,t+1} = (1+g^)A_{0,t}$ and $L_{t+1} = (1+n)L_t$ yields*

$$K_{t+1} = k^* A_{0,t+1} L_{t+1} = k^* (1+g^*)(1+n) A_{0,t} L_t,$$

so $K_{t+1}/K_t = (1+g^)(1+n)$. The same argument applies to H_t . Substituting $K_t = k^* A_{0,t} L_t$ and $H_t = h^* A_{0,t} L_t$ into the production function yields*

$$Y_t = (k^*)^\alpha (h^*)^\beta \Omega(s^*)^{1-\alpha-\beta} A_{0,t} L_t,$$

so $Y_{t+1}/Y_t = (1+g^)(1+n)$ and $y_t = Y_t/(A_{0,t}L_t)$ is constant at y^* . $\square$*

8.2 Existence of BGP

Lemma 8.2 (BGP Existence). *Suppose:*

1. $\Omega(s)$ and $\Gamma(h)$ are continuous and bounded on their domains
2. $g(s) \in [g_0, g_{\max}]$ is bounded
3. $\theta_H(s) \in [\theta_{\min}, \theta_{\max}]$ is bounded

Then the map $(k, h) \mapsto (k', h')$ defined by the laws of motion is continuous and maps a compact convex set into itself. By Brouwer's fixed-point theorem, a BGP (k^, h^*) exists.*

Proof: See Appendix F.

8.3 Steady-State Characterization

Setting $k_{t+1} = k^*$ and $h_{t+1} = h^*$:

$$\lambda_K(g^*)k^* = \theta_K^* \varsigma y^* \quad (48)$$

$$\lambda_H(g^*)h^* = \theta_H^* \varsigma y^* \quad (49)$$

where:

$$\lambda_K(g) = g + n + \delta_K + gn, \quad \lambda_H(g) = g + n + \delta_H + gn \quad (50)$$

Dividing:

$$\phi^* \equiv \frac{h^*}{k^*} = \frac{\theta_H^*}{\theta_K^*} \cdot \frac{\lambda_K(g^*)}{\lambda_H(g^*)} \quad (51)$$

From the first equation:

$$k^* = \left[\frac{\theta_K^* \varsigma (\phi^*)^\beta (\Omega^*)^{1-\alpha-\beta}}{\lambda_K(g^*)} \right]^{\frac{1}{1-\alpha-\beta}} \quad (52)$$

$$h^* = \phi^* k^* \quad (53)$$

where:

- $\theta_H^* = \bar{\theta} + \eta(\pi - s^*)$
- $\theta_K^* = 1 - \theta_H^*$
- $\Omega^* = [s^* \omega_U + (1 - s^*)]^\psi$
- $g^* = g_0 + \phi(s^*)^\zeta$
- $s^* = \Gamma(h^*)$

Fixed-point system:

$$s^* = \Gamma(h^*(s^*)) \quad (54)$$

$$g^* = g_0 + \phi(s^*)^\zeta \quad (55)$$

8.4 Numerical Solution Algorithm

Algorithm 8.1 (Steady-State Solver). 1. **Initialize:** Guess $s^{(0)} \in (0, \pi)$

2. **Compute growth rate:** $g^{(j)} = g_0 + \phi(s^{(j)})^\zeta$

3. **Compute effective TFP:** $\Omega^{(j)} = [s^{(j)} \omega_U + (1 - s^{(j)})]^\psi$

4. **Compute investment allocation:**

- $\theta_H^{(j)} = \bar{\theta} + \eta(\pi - s^{(j)})$
- $\theta_K^{(j)} = 1 - \theta_H^{(j)}$

5. **Compute effective depreciation rates:**

- $\lambda_K^{(j)} = g^{(j)} + n + \delta_K + g^{(j)}n$
- $\lambda_H^{(j)} = g^{(j)} + n + \delta_H + g^{(j)}n$

6. **Compute capital ratio:** $\phi^{(j)} = \frac{\theta_H^{(j)} \lambda_K^{(j)}}{\theta_K^{(j)} \lambda_H^{(j)}}$

7. Compute steady-state capital:

- $k^{(j)} = \left[\frac{\theta_K^{(j)} \varsigma(\phi^{(j)})^\beta (\Omega^{(j)})^{1-\alpha-\beta}}{\lambda_K^{(j)}} \right]^{\frac{1}{1-\alpha-\beta}}$
- $h^{(j)} = \phi^{(j)} k^{(j)}$

8. Solve implicit equation: Find $s^{(j+1)} \in [0, \pi]$ satisfying:

$$(s^{(j+1)})^{1/(1-\mu)} (\pi - s^{(j+1)}) = \tilde{\chi}(h^{(j)})^{\mu/(1-\mu)}$$

using Newton-Raphson or bisection.

9. Check convergence: If $|s^{(j+1)} - s^{(j)}| < \varepsilon$ (e.g., $\varepsilon = 10^{-6}$), stop. Otherwise, set $j \leftarrow j+1$ and return to step 2.

8.5 Multiplicity of BGPs

Proposition 8.1 (Uniqueness Condition). *If $\zeta < 1$ (diminishing returns in innovation) and η is sufficiently small, the BGP is unique.*

Proposition 8.2 (Multiple BGPs). *If $\zeta > 1$ (increasing returns in innovation) and η is large, multiple BGPs may exist:*

- **Low-growth trap:** Low urbanization $s_L^* \rightarrow$ low innovation $g_L^* \rightarrow$ low housing investment $\rightarrow$ reinforces low urbanization
- **High-growth equilibrium:** High urbanization $s_H^* \rightarrow$ high innovation $g_H^* \rightarrow$ high housing investment $\rightarrow$ reinforces high urbanization

Analysis note: Numerical bifurcation analysis will be conducted once parameters are calibrated. See **Appendix G** for the theoretical framework.

9 LOCAL STABILITY ANALYSIS

9.1 Linearization

The dynamics near the BGP are governed by the Jacobian matrix:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial k_{t+1}}{\partial k_t} & \frac{\partial k_{t+1}}{\partial h_t} \\ \frac{\partial h_{t+1}}{\partial k_t} & \frac{\partial h_{t+1}}{\partial h_t} \end{bmatrix} \Big|_{(k^*, h^*)} \quad (56)$$

Proposition 9.1 (Stability Condition). *The BGP (k^*, h^*) is locally asymptotically stable if both eigenvalues of $\mathbf{J}$ satisfy $|\lambda_i| < 1$.*

Sign pattern (qualitative analysis):

- $\frac{\partial k'}{\partial k} > 0$: Higher business capital today $\rightarrow$ higher business capital tomorrow (persistence)
- $\frac{\partial h'}{\partial h} > 0$: Higher housing capital today $\rightarrow$ higher housing capital tomorrow (persistence)
- $\frac{\partial k'}{\partial h}$: Ambiguous sign
 - **Positive effect:** More housing $\rightarrow$ higher $s \rightarrow$ higher $\Omega \rightarrow$ higher output $\rightarrow$ higher k'
 - **Negative effect:** More housing $\rightarrow$ higher $\theta_H \rightarrow$ lower $\theta_K \rightarrow$ lower k'

- **Growth effect:** More housing $\rightarrow$ higher $s \rightarrow$ higher $g(s) \rightarrow$ affects denominator
- $\frac{\partial h'}{\partial k}$: Typically positive (higher $k \rightarrow$ higher $y \rightarrow$ higher h')

Analysis note: Numerical evaluation of eigenvalues will be performed once parameters are calibrated. The full Jacobian derivation and parameter sensitivity analysis are provided in **Appendix H**.

10 CONTRIBUTION AND RELATIONSHIP TO LITERATURE

10.1 Core Contributions

This model makes three novel contributions:

1. **Housing bottlenecks affect access to the idea engine of cities:** By linking housing frictions to urban occupancy s_t and innovation $g(s_t)$, the model shows how housing shortages reduce long-run growth—not just through lower TFP today, but through slower frontier advancement.
2. **Tight, coherent spatial block with matching frictions and quality scaling:** The combination of search-and-matching frictions, endogenous vacancy rates, and quality-adjusted housing creates a microfounded general-equilibrium block that is fully consistent with balanced growth. This resolves the “knife-edge” problem in previous spatial growth models.
3. **Policy-relevant long-run growth implications:** The endogenous growth channel $g(s)$ creates a new margin for housing policy: reforms that increase urban density raise both current productivity and the rate of frontier innovation, generating compounding welfare gains.

10.2 Relationship to Existing Literature

Spatial economics:

- **Duranton & Puga (2004):** Our model extends their “basic vs. advanced” city framework by adding capital accumulation and endogenous growth. While they focus on static spatial sorting, we analyze dynamic capital allocation and long-run growth consequences.
- **Desmet & Rossi-Hansberg (2014):** We share the focus on spatial productivity but add housing market frictions and explicit two-capital dynamics. Their quantitative framework is complementary to our analytically tractable growth model.
- **Rossi-Hansberg et al. (2019):** We complement their quantitative spatial model with an analytically tractable growth framework that isolates the housing-innovation channel.

Innovation and agglomeration:

- **Bloom et al. (2020):** We microfound their empirical finding that innovation is geographically concentrated by linking housing supply to urban density and growth.
- **Jones (1995):** Our $g(s)$ function embeds semi-endogenous growth with location-specific R&D. We add the housing constraint as a barrier to accessing the R&D-intensive urban sector.
- **Moretti (2004):** Our agglomeration function $\Omega(s)$ captures the human capital externalities he documents empirically. We extend this to show how housing frictions limit access to these externalities.

Housing economics:

- **Glaeser & Gyourko (2005):** We microfound their observation that housing supply constraints limit city growth by embedding it in a two-capital growth model with endogenous innovation.
- **Hsieh & Moretti (2019):** Our model provides a growth-theoretic framework for quantifying the aggregate costs of their documented spatial misallocation.

Two-capital growth models:

- **Davis & Heathcote (2005):** We extend the standard Solow-Swan model with housing by adding spatial heterogeneity, matching frictions, and endogenous growth. Their focus is on cyclical dynamics; ours is on long-run growth.
- **Rebelo (1991):** We extend AK-style endogenous growth to spatial settings with housing constraints.

Key distinction: This is the first model to combine:

- Search-and-matching frictions in housing markets
- Quality-adjusted housing capital consistent with balanced growth
- Spatial productivity differences with agglomeration
- Endogenous frontier growth through urban innovation
- Tractable two-capital dynamics

in a unified analytical framework.

10.3 Comparative Statics: Housing Frictions and Long-Run Growth

This subsection states the key comparative statics that link structural housing frictions to the long-run growth rate. The formal proofs, which rely on the implicit function theorem and the monotonicity properties of $\Gamma(h)$ and $g(s)$, are delegated to a future Appendix J.

We focus on parameters that loosen or tighten the effective housing constraint:

- π : share of households that are unconstrained in accessing urban housing
- χ : matching efficiency in the housing market
- σ : separation rate (churn) of occupied units
- δ_v : depreciation of vacant units
- η : responsiveness of the investment share in housing to congestion

Let $g^*(\pi, \chi, \sigma, \delta_v, \eta)$ denote the BGP frontier growth rate associated with the unique BGP (k^*, h^*) and urbanization level $s^* = \Gamma(h^*)$.

Assumption 10.1. For the following:

1. The functions $\Gamma(h)$, $\Omega(s)$ and $g(s)$ are continuously differentiable and strictly monotone in their arguments:

$$\Gamma'(h) > 0, \quad \Omega'(s) > 0, \quad g'(s) > 0.$$

2. For the parameter range considered, the dynamical system in (k_t, h_t) admits a unique BGP (k^*, h^*) and this BGP is locally stable.
3. The reduced-form housing investment share $\theta_H(s)$ is continuously differentiable and locally strictly increasing in congestion $(\pi - s)$, i.e.

$$\frac{\partial \theta_H(s)}{\partial (\pi - s)} > 0.$$

Theorem 10.1 (Comparative Statics of Growth with Respect to Housing Frictions). *Under Assumption 10.1, consider a marginal change in the structural parameters $(\pi, \chi, \sigma, \delta_v, \eta)$ around a baseline BGP with growth rate g^* . Then, to first order:*

1. *Loosening access to urban housing increases long-run growth:*

$$\frac{\partial g^*}{\partial \pi} > 0, \quad \frac{\partial g^*}{\partial \chi} > 0.$$

2. *Intensifying housing churn or vacancy decay reduces long-run growth:*

$$\frac{\partial g^*}{\partial \sigma} < 0, \quad \frac{\partial g^*}{\partial \delta_v} < 0.$$

3. *Stronger investment responsiveness amplifies the growth impact of constraints:** For η sufficiently small,*

$$\frac{\partial g^*}{\partial \eta} > 0$$

if and only if the indirect effect of higher housing investment on innovation (via s^) dominates its crowding-out effect on business capital k^* .*

***Sketch of intuition:** The BGP growth rate is $g^* = g(s^*)$, with $s^* = \Gamma(h^*)$ and h^* determined by the steady-state capital accumulation conditions. An increase in π or χ shifts the implicit occupancy equation

$$s^{1/(1-\mu)}(\pi - s) = \tilde{\chi} h^{\mu/(1-\mu)}$$

outward, raising s^* for given h^* , while higher σ or δ_v shifts it inward. By monotonicity of Γ , s^* moves in the same direction, and by monotonicity of g we obtain the signs in parts (1)–(2). The effect of η operates through the investment allocation rule: a higher η intensifies the response of θ_H to congestion, which raises h^* and s^* but also reduces k^* via the resource constraint. When η is small, the first-order effect via s^* dominates, so g^* rises; for larger η , the sign becomes ambiguous and multiple BGPs can arise, as discussed in Propositions 8.1–8.2.

A formal statement and proof of Theorem 10.1, using the implicit function theorem on the stacked steady-state system, will be developed in a future Appendix J. Here we emphasize the economic content: policies that relax structural housing constraints (higher π and χ , lower σ and δ_v) raise the long-run frontier growth rate in this model.

NOTATION TABLE

Symbol	Definition	Domain
t	Time period	$\{0, 1, 2, \dots\}$
L_t	Population	$\mathbb{R}_{++}$
n	Population growth rate	$[0, 0.05]$
$A_{0,t}$	Frontier technology	$\mathbb{R}_{++}$
$g(s_t)$	Endogenous growth rate	$[g_0, g_{\max}]$
g_0	Baseline growth rate	$\mathbb{R}_+$
ϕ	Innovation elasticity	$\mathbb{R}_+$
ζ	Growth curvature	$\mathbb{R}_+$
s_t	Urban occupancy share	$[0, \pi]$
π	Unconstrained household share	$(0, 1)$
K_t	Business capital	$\mathbb{R}_+$
H_t	Quality-adjusted housing capital	$\mathbb{R}_+$
$\tilde{H}_t$	Physical housing stock	$\mathbb{R}_+$
q_t	Average housing quality	$\mathbb{R}_{++}$
κ	Quality scaling constant	$\mathbb{R}_{++}$
Y_t	Aggregate output	$\mathbb{R}_+$
α	Business capital share	$(0, 1)$
β	Housing capital share	$(0, 1)$
$A_{\text{eff},t}$	Effective TFP	$\mathbb{R}_{++}$
$\Omega(s)$	Agglomeration function	$[1, (\pi\omega_U + (1 - \pi))^\psi]$
ω_U	Urban productivity	$(1, \infty)$
ω_R	Rural productivity (normalized)	$\{1\}$
ψ	Agglomeration elasticity	$\mathbb{R}_+$
H_t^{occ}	Occupied housing	$[0, H_t]$
H_t^{vac}	Vacant housing	$[0, H_t]$
B_t	Searching households	$[0, \pi L_t]$
M_t	Housing matches	$[0, \min\{H_t^{\text{vac}}, B_t\}]$
χ	Matching efficiency	$\mathbb{R}_{++}$
μ	Matching elasticity	$(0, 1)$
σ	Separation rate	$(0, 1)$
δ_v	Vacancy depreciation	$(0, 1)$
v^*	Steady-state vacancy rate	$(0, 1)$
$\Gamma(h)$	Occupancy function	$[0, \pi]$
$\tilde{\chi}$	Reduced-form matching parameter	$\mathbb{R}_{++}$
$r_{K,t}$	Business capital rental rate	$\mathbb{R}_+$
$r_{H,t}$	Housing rental rate	$\mathbb{R}_+$
ρ_t	Congestion rent	$\mathbb{R}_+$
ς	Saving rate	$(0, 1)$
$\theta_{H,t}$	Housing investment share	$[\theta_{\min}, \theta_{\max}]$
$\theta_{K,t}$	Business investment share	$[1 - \theta_{\max}, 1 - \theta_{\min}]$
$\bar{\theta}$	Baseline housing share	$(0, 1)$
η	Policy responsiveness	$\mathbb{R}_+$
φ	Adjustment cost parameter	$\mathbb{R}_{++}$
ρ_1	Congestion rent slope	$\mathbb{R}_+$
δ_K	Business capital depreciation	$(0, 1)$
δ_H	Housing capital depreciation	$(0, 1)$
k_t	Business capital (intensive)	$\mathbb{R}_+$
h_t	Housing capital (intensive)	$\mathbb{R}_+$
y_t	Output (intensive)	$\mathbb{R}_+$